

AI College Recommendation Expert System

A Rule-Based Expert System Using Prolog

Project Report

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Technology: SWI-Prolog

Domain: Artificial Intelligence / Expert Systems

Abstract

Selecting a suitable college is a complex decision-making process that involves evaluating multiple factors such as entrance examination scores, branch preferences, college rankings, placement statistics, fee structures, scholarship opportunities, and hostel facilities. Students often face difficulty in comparing institutions and identifying colleges that best match their academic profile and personal requirements.

The AI College Recommendation Expert System addresses this problem through a rule-based approach implemented in Prolog. The system utilizes a knowledge base containing detailed information about colleges and applies inference rules to analyze student profiles. Based on eligibility criteria, scoring mechanisms, and decision-making rules, the system generates personalized college recommendations.

The project demonstrates the application of Artificial Intelligence concepts such as knowledge representation, rule-based reasoning, inference mechanisms, and expert systems in solving real-world educational decision-support problems.

1. Introduction

Artificial Intelligence (AI) enables computer systems to mimic human reasoning and decision-making processes. One of the earliest and most successful applications of AI is the Expert System.

An Expert System is a computer program that uses domain-specific knowledge and inference rules to solve problems that typically require human expertise. In the field of education, students often seek guidance regarding college selection. Educational counselors usually evaluate a student's examination performance, financial constraints, preferred branch, and career goals before suggesting suitable institutions.

The AI College Recommendation Expert System replicates this decision-making process using Prolog. The system stores knowledge about colleges and applies logical rules to generate recommendations that align with a student's profile.

2. Objectives

The primary objectives of this project are:

- To develop a rule-based college recommendation system.
- To implement expert system concepts using Prolog.
- To provide personalized college recommendations.
- To perform eligibility checking based on entrance examination scores.
- To classify colleges into different tiers.
- To compare colleges on multiple parameters.
- To assist students in making informed educational decisions.
- To demonstrate knowledge representation and inference mechanisms.

3. System Architecture

The system follows a layered architecture consisting of multiple components.

Student Profile Input



Eligibility Engine



Knowledge Base



Scoring Engine



Recommendation Engine



Top College Recommendations

The Student Profile Input collects information such as examination scores, preferred branch, budget, location preference, and hostel requirements.

The Eligibility Engine filters colleges based on admission criteria.

The Knowledge Base stores detailed information regarding colleges, rankings, placements, fees, scholarships, and facilities.

The Scoring Engine evaluates colleges using multiple parameters and assigns scores.

The Recommendation Engine generates ranked recommendations for the student.

(Insert system_architecture.png here)

4. Technologies Used

The project was developed using the following technologies:

- SWI-Prolog
 - Artificial Intelligence
 - Expert Systems
 - Knowledge Representation
 - Logic Programming
 - Rule-Based Reasoning
 - Decision Support Systems
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5. Methodology

The system operates through the following stages:

Step 1: Student Profile Collection

The user provides:

- Entrance examination score
- Preferred course
- Preferred branch
- Budget
- Preferred state
- Hostel requirement

Step 2: Eligibility Verification

The system checks whether the student satisfies the admission criteria of each college.

Step 3: Knowledge Base Querying

Relevant college information is retrieved from the knowledge base.

Step 4: Score Computation

Each college is evaluated based on:

- NIRF ranking
- Placement statistics
- Fee affordability
- Accreditation
- Location preference

Step 5: Recommendation Generation

The recommendation engine generates ranked college suggestions based on computed scores.

Step 6: Advisory Support

Additional advisory rules identify:

- Safe colleges
 - Stretch colleges
 - Dream colleges
 - Scholarship opportunities
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6. Key Features

The system provides the following features:

- Personalized college recommendations
- Eligibility verification
- Tier classification
- College comparison
- Placement analysis
- Scholarship analysis
- Budget-aware recommendations
- Hostel availability checking

- Safe college suggestions
 - Stretch college identification
 - Rule-based decision support
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7. Sample Outputs

The following functionalities were successfully implemented and tested:

Top Recommendations

(Insert top_recommendations.png)

College Profile

(Insert college_profile.png)

College Comparison

(Insert college_comparison.png)

Tier Classification

(Insert tier_classification.png)

Safe College Suggestions

(Insert safe_colleges.png)

8. Future Scope

Several enhancements can be incorporated in future versions:

- Web-based graphical user interface
 - Dynamic college database integration
 - Cloud deployment
 - Mobile application support
 - Real-time ranking updates
 - Machine Learning assisted recommendations
 - Student feedback analysis
 - Personalized career guidance modules
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9. Conclusion

The AI College Recommendation Expert System successfully demonstrates the practical application of Artificial Intelligence concepts through a rule-based expert system. The project integrates knowledge representation, inference mechanisms, eligibility checking, and scoring algorithms to provide meaningful recommendations to students.

The modular architecture, recommendation engine, advisory rules, and decision-support capabilities make the system an effective educational guidance tool. The project highlights how Prolog and expert systems can be used to solve real-world decision-making problems in an efficient and interpretable manner.

The system can serve as a foundation for future intelligent educational advisory platforms and demonstrates strong concepts of Artificial Intelligence, Logic Programming, and Software Engineering.